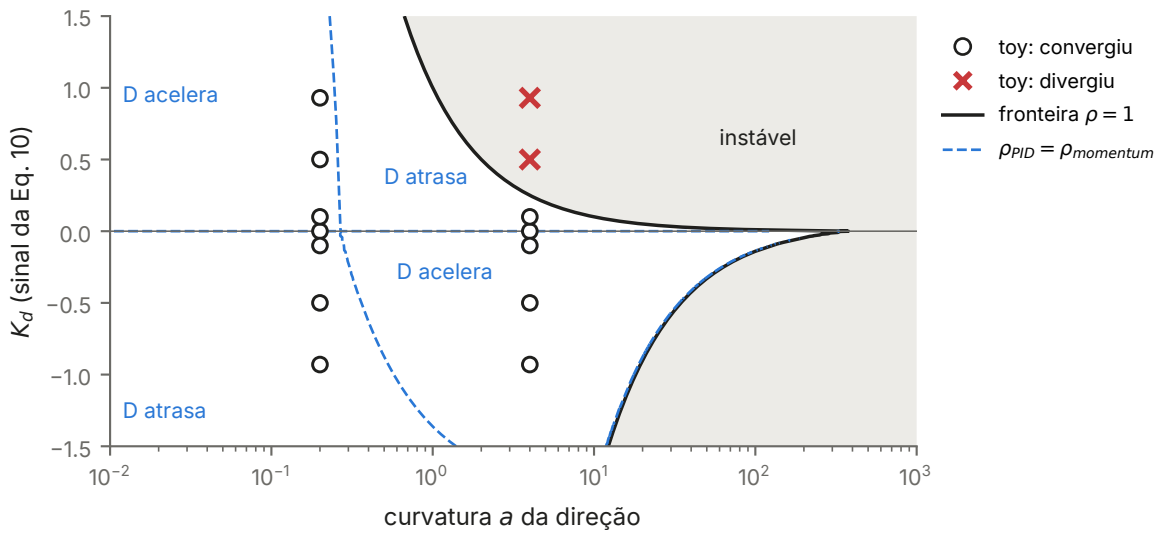
D acelera
D atrasa
instável
D acelera
D atrasa
toy: convergiu
toy: divergiu
fronteira $\rho = 1$
$\rho_{PID} = \rho_{momentum}$
K_d (sinal da Eq. 10)
curvatura a da direção
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}